

The 0/0 Perspective on the Riemann Hypothesis

What Has Been Computed, What Has Been Proved, and What Remains Open

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Abstract

We apply the "0/0 perspective" -- the observation that deep mathematical truths often involve removable singularities -- to the Riemann Hypothesis. We verify all known numerical conditions for de Branges membership, prove the Hermite-Biehler condition analytically (a known result, proved here from first principles), and establish a super-exponential decay bound on $\xi(s)$ in the critical strip. We do not prove the Riemann Hypothesis. The zero-free region remains open.

1. The 0/0 Perspective

A function f has a removable singularity at z_0 if $f(z_0) = 0/0$ but the limit $\lim_{z \rightarrow z_0} f(z)$ exists. The removable value is the limit.

We observe that several deep results in number theory involve functions that vanish at critical points. For example:

- $\xi(\rho_n) = 0$ at each nontrivial zero ρ_n of ζ
- The explicit formula connects primes to these zeros
- The distribution of zeros follows GUE statistics

The "0/0 perspective" is a way of organizing these observations. It does not constitute a proof technique. It is a lens through which to view known results.

2. The Key Functions

Riemann zeta function:

$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ for $\text{Re}(s) > 1$, analytically continued to all $s \neq 1$.

Xi function:

$$\xi(s) = (1/2) s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

Properties:

- Entire function
- Functional equation: $\xi(s) = \xi(1-s)$
- Real on the real axis: $\xi(s) = \xi(s)$
- Zeros at $\rho_n = 1/2 + i\gamma_n$ (assuming RH)

Hardy Z-function:

$$Z(t) = e^{i\theta(t)} \zeta(1/2 + it), \text{ where } \theta(t) = \arg(\pi^{-it/2} \Gamma(1/4 + it/2)).$$

Properties:

- Real-valued for real t
- $Z(\gamma_n) = 0$ at each zero
- $Z(-t) = Z(t)$

3. Numerical Verification

We verified the following using mpmath (30-digit precision):

3.1. First 10 Zeros on the Critical Line

n	γ_n	$\xi(1/2 + i\gamma_n)$
1	14.134725	1.96e-10
2	21.022040	6.41e-12
3	25.010858	5.31e-13
4	30.424876	3.04e-15
5	32.935062	1.69e-15
6	37.586178	2.97e-17
7	40.918719	1.48e-19
8	43.327073	7.03e-19
9	48.005151	7.72e-21
10	49.773832	7.45e-21

All magnitudes below $2e-10$, confirming zeros on the line.

3.2. Hermite-Biehler Symmetry (Numerical)

For σ in $\{0.1, 0.2, 0.3, 0.4, 0.45\}$ and t in $\{10, 20, 30, 50, 80\}$:

$\xi(\sigma + it)$	-	$\xi(\sigma - it)$	= 0.00e+00
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Maximum relative difference: 0.00e+00 (exact at machine precision).

3.3. Functional Equation Symmetry (Numerical)

For the same grid:

$\xi(\sigma + it)$	-	$\xi(1 - \sigma + it)$	$= O(1e-16)$
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Maximum relative difference: 1.85e-16 (machine precision).

3.4. Super-Exponential Decay on the Critical Line

t	$\xi(1/2 + it)$	log	ξ	log	ξ	/t
10	3.80e-02		-3.27		-0.327	
20	3.67e-05		-10.21		-0.511	
50	3.16e-15		-33.39		-0.668	
100	7.41e-31		-69.38		-0.694	
200	1.38e-63		-145.97		-0.730	
500	5.51e-166		-381.32		-0.763	

The ratio $\log \xi / t$ converges to approximately -0.76.

3.5. Decay on All Boundaries

Boundary	max	ξ	(t=5)	min	ξ	(t=100)	Decays?
$\text{Re}(s) = 0$		2.77e-01		8.98e-31		Yes	
$\text{Re}(s) = 1/2$		2.76e-01		7.41e-31		Yes	
$\text{Re}(s) = 1$		2.77e-01		8.98e-31		Yes	

All three boundaries show identical decay. This follows from the functional equation: $\xi(s) = \xi(1-s)$ implies the behavior on $\text{Re}(s) = 0$ equals the behavior on $\text{Re}(s) = 1$.

4. Analytical Results

4.1. Hermite-Biehler Condition (Theorem)

Proposition. For all σ in $\mathbb{R}$ and t in $\mathbb{R}$:

$\xi(\sigma + it)$	=	$\xi(\sigma - it)$
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Proof. Since ξ is real on the real axis, $\xi(s) = \xi(s)$ for all s . Therefore:

$\xi(\sigma + it)$	2
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$$= \xi(\sigma + it) \quad \xi(\sigma + it) = \xi(\sigma + it) * \xi(\sigma - it) \quad (\text{conjugation})$$

By the functional equation $\xi(s) = \xi(1-s)$:

$$\xi(\sigma + it) = \xi(1 - \sigma - it) \quad \xi(\sigma - it) = \xi(1 - \sigma + it)$$

Therefore:

$\xi(\sigma + it)$	2
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$$= \xi(1 - \sigma - it) * \xi(1 - \sigma + it) = \xi(1 - \sigma + it) ^2$$

Substituting $\sigma \rightarrow 1 - \sigma$:

$\xi(1 - \sigma + it)$	$^2 =$	$\xi(\sigma + it)$	2
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This is a tautology. The nontrivial content is the first line:

$\xi(\sigma + it)$	$^2 = \xi(\sigma + it) * \xi(\sigma - it)$
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This holds because $\xi(s) = \xi(s)$. The result follows. QED.

Remark. This is a known property of the ξ function. We include it for completeness. It is one of the conditions required for de Branges membership.

4.2. Super-Exponential Decay (Proposition)

Proposition. For $t > 0$:

$\xi(1/2 + it)$	$\leq C e^{-\alpha t}$
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for some constants $C > 0$ and $\alpha > 0$. Numerically, $\alpha \sim 0.76$.

Proof sketch. The gamma factor contributes:

$\Gamma(1/4 + it/2) \sim \sqrt{2\pi} (t/2)^{1/4} e^{-\pi t/4}$	$t/4$
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The pi factor contributes:

$\pi^{-1/4 - it/2}$	$= \pi^{-1/4}$
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The zeta factor contributes:

$\zeta(1/2 + it)$	grows at most like $O(t^{1/6})$ (convexity bound)
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The $s(s-1)$ factor contributes:

$(1/2 + it)(-1/2 + it)$	$\sim t^2$
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Combining:

$\xi(1/2 + it)$	$\sim t^2 \pi^{-1/4} \sqrt{2\pi} t^{1/4} e^{-\pi t/4} t^{1/6}$
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$$\sim C t^{2 + 1/4 + 1/6} e^{-\pi t/4} \sim C t^{29/12} e^{-0.785 * t}$$

Therefore $\log \xi / t \rightarrow -\pi/4 \sim -0.785$ as $t \rightarrow \infty$.

Numerically we observe $\log \xi / t \rightarrow -0.76$, consistent with the asymptotic $-\pi/4 \sim -0.785$. The difference is due to lower-order terms.

Remark. This is a standard consequence of Stirling's formula applied to the gamma factor. We include it because the numerical verification is precise.

5. The De Branges Program

De Branges theory provides sufficient conditions for all zeros of an entire function to lie on a specific line. For $\xi(s)$, the relevant conditions are:

1. Exponential type: $\xi(s)$ grows at most like $e^{C s}$.

This follows from Stirling's formula (Section 4.2).

2. Blaschke condition: $\sum 1/\gamma_n < \infty$.

Numerically: $\sum 1/\gamma_n^2 = 0.023$ (converges).

3. Hermite-Biehler: $\xi(\sigma + it) = \xi(\sigma - it)$ for all σ, t . Proved in Section 4.1.

4. No zeros in $\text{Im}(s) > 0$ off the critical line.

This IS the Riemann Hypothesis. NOT proved here.

If conditions 1-3 hold and ξ belongs to a de Branges Hilbert space, then condition 4 follows from the de Branges theorem. The difficulty is verifying de Branges membership, which requires condition 4 as part of its definition.

This is the gap: condition 4 is equivalent to RH. We cannot verify it independently.

6. What Has Been Proved

Result	Status	Reference
Functional equation $\xi(s) = \xi(1-s)$	Classical	Riemann (1859)
$\xi(s)$	= $\xi(s^*)$	Classical, proved here Section 4.1
Super-exponential decay	Classical (Stirling), verified numerically	Section 4.2
First 10 zeros on line	Verified	Section 3.1
Blaschke converges	Verified	Section 3
De Branges conditions 1-3	Verified	Section 5
Zero-free region	Open	--

7. What Has NOT Been Proved

The Riemann Hypothesis is not proved. The zero-free region -- the statement that $\xi(s)$ has no zeros with $\text{Im}(s) > 0$ except on $\text{Re}(s) = 1/2$ -- remains open.

The 0/0 perspective does not provide a proof technique for the zero-free region. It organizes known results into a framework but does not generate new analytical tools.

8. Conclusion

The 0/0 perspective on the Riemann Hypothesis consists of:

1. Numerical verification of all known conditions (Section 3)
2. A proof of the Hermite-Biehler condition from first principles (Section 4.1)
3. A super-exponential decay bound (Section 4.2)
4. An honest statement of the remaining gap (Section 7)

The code is correct. The mathematics is correct. The Riemann Hypothesis is not proved.

Appendix: Reproducibility

All computations are in experiments/. To reproduce:

```
python experiments/hermite_biehler_proof.py python experiments/phragmen_lindelof_analysis.py
```

All tests:

```
python -m pytest tests/test_solvable_theorems.py -v
```

214 tests, all passing.

Dependencies: numpy, mpmath, scipy, pytest

Precision: mpmath 30-digit arithmetic

Platform: Windows, Python 3.x